

Sahil Joshi

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Education

Rice University

Houston, TX

Doctor of Philosophy in Computer Science

Jan 2025 – Present

- Research: Randomized Algorithms; Large Scale Machine Learning; Sketching
- Coursework: Deep Learning Systems, Probabilistic Data Structures and Algorithms, LLM Fine-Tuning (Vizura), Harness Engineering (Vizura), IoT Programming and Data Analysis

University of Massachusetts, Amherst

Amherst, MA

Bachelor of Science in Computer Science and Mathematics

Sept 2019 – May 2023

- GPA: 4.0; Chancellor's Scholarship; Baystate Fellowship
- Outstanding Undergraduate Award; Dean's List; Outstanding Undergraduate Course Assistant Award (2021)
- Coursework: Machine Learning, Artificial Intelligence, Introduction to Algorithms, Information Systems
- Honors Thesis: Detection of Distributed Denial Of Service Attacks Using Binary Signals

Professional Experience

Rush Lab

Houston, TX

Researcher (Advisor: Professor Anshumali Shrivastava)

June 2025 – Present

- Developed a novel linear-time, linear-memory attention mechanism using Repeated Arrays-of-Count Estimators (RACE), enabling efficient scaling to extremely large context windows while maintaining competitive performance with baselines on publicly available benchmarks.
- Proposed a novel soft Locality Sensitive Hashing (LSH) formulation that outperforms traditional LSH on retrieval tasks, enabling practical data-agnostic sparse attention with high-performance Triton kernels.

Workato, Inc.

Palo Alto, CA

AI Engineering Intern

June 2026 – Aug 2026

- Researched coding agents, evaluation harnesses, security guardrails, and autonomous software engineering workflows for enterprise automation; designed and built an agentic harness from the ground up.
- Fine-tuned 1B-parameter Large Language Models (LLMs) and systematically studied the effects of training data, hyperparameters, and model scale on performance.
- Led adversarial red-teaming of OpenAI models using fine-tuned LLMs to evaluate safety and security controls for enterprise deployment.

Dell Technologies

Round Rock, TX

Software Engineer 1 (DevOps Engineer)

July 2023 – Jan 2025

- Enhanced the user interface and optimized backend services of the Pipeline Dashboard application, used for managing Object Scale components similar to AWS S3.
- Implemented CI/CD automation using Jenkins and GitLab for deployment pipelines, improving release efficiency and reducing manual intervention across Object Scale services.
- Integrated observability tools (Prometheus, Grafana) to monitor service health, enabling proactive issue detection and reducing downtime.

Lutron Electronics

Philadelphia, PA

Software Engineer Co-Op (iOS App Developer)

May 2022 – Dec 2022

- Improved the design and workflow for the Lutron app, which is used for controlling lights, fans and shades. Fixed bugs on a bi-weekly basis to improve the iOS app's ratings.
- Designed the feature for Widget Zone Control. This feature lets the users control individual devices from the widget by providing them more flexibility from outside the app.
- Partnered with other students to think about test strategies for the new feature and wrote unit tests to minimize the errors in the app.

Publications

S. Joshi, A. Chowdhury, A. Kanakamedala, E. Singh, E. Tu, A. Shrivastava. *RACE Attention: An Inherently Efficient Linear-Time Attention Mechanism for Long Sequences*. ICLR 2026.

H. Le, **S. Joshi**, Z. Yang, Z. Xu, A. Shrivastava. *Scout Before You Attend: Sketch-and-Walk Sparse Attention for Efficient LLM Inference*. ICML 2026.

S. Joshi, P. Dixit, A. Chowdhury, A. Shrivastava, J. E. Gonzalez, I. Stoica, K. K. Agrawal, A. Desai. *Inference Time Context Sparsity: Illusion or Opportunity?* arXiv preprint, 2026.

S. Joshi, A. Chowdhury, W. Bellinger, A. Kanakamedala, E. Singh, H. Le, A. Desai, A. Shrivastava. *SOCKET: SOft Collison Kernel EsTimator for Sparse Attention*. arXiv preprint, 2026.

Open-Source Projects

Sparse Attention Hub

Core contributor to an end-to-end sparse attention inference infrastructure for off-the-shelf Large Language Models, enabling rapid experimentation and fair, apples-to-apples evaluation of decode-time methods.

OMI: Metagenomics Assistant

Developed an LLM-powered assistant that uses Retrieval-Augmented Generation (RAG) to translate natural language prompts into YAML configurations for a custom Nextflow pipeline, enabling non-experts to run long-read metagenomic analyses with automated tool selection, parameter tuning, and workflow documentation.

Heritage Metadata Extractor

Built an end-to-end prototype using OCR tools and (LLMs) to digitize handwritten heritage documents, extract key metadata, store it in a structured database, and support semantic search through natural language queries.

Teaching

Teaching Assistant

Rice University

IoT Programming and Data Analysis, Probabilistic Algorithms and Data Structures

Spring – Fall 2026

- Held weekly office hours to support students with course concepts, programming assignments, and project debugging; assisted with grading and feedback.

Research Mentor

Rice University

Mentoring Undergraduate Researchers

Summer 2025 – Present

- Mentoring 4 undergraduate researchers (Amar Kanakamedala, Ekam Singh, Wyatt Bellinger, Evan Tu) on projects in efficient training/inference of large language models and sparse attention.
- Guiding students through problem formulation, literature review, and experimentation, leading to independent project development and open-source contributions.

Activities

Rice University Computer Science Graduate Student Association

Houston, TX

President

June 2025 – Present

- Organized seminar series by inviting computer science faculty from Texas and neighboring states to present research to graduate students.
- Led initiatives to strengthen graduate student support, including launching Teaching Assistant Awards, organizing community events, and enabling travel funding for master's students.

Workshops and Conferences

Reviewer

May 2026 – Sep 2026

- NeurIPS 2026

Programming Skills

Languages: Python, Triton, CUDA, Java, C++, SQL, $\text{\LaTeX}$.

Frameworks: PyTorch, NumPy, Pandas.